

Benoît Bonnet-Weill

Chargé de Recherche CNRS

Junior CNRS Researcher

(Last updated on February 2, 2025)

Personal information

Civil Status: Born the 27th of April 1993 in Paris, XII^{ième} arrondissement.

Married since the 25th of August 2018, one lovely daughter.

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Google Scholar: <https://scholar.google.fr/citations?user=0w5eQawAAAAJ&hl=fr>

Academic positions

- ◇ **January 2024 – now:** Junior CNRS Researcher in the team MODÉLISATION, ESTIMATION ET ANALYSE DES SYSTÈMES (MODESTY), *Laboratoire des Signaux et Systèmes*, Gif-sur-Yvette.
- ◇ **November 2021 – December 2023:** Junior CNRS Researcher in the team MÉTHODES ET ALGORITHMES DE COMMANDE (MAC), *Laboratoire d'Analyse et d'Architecture des Systèmes*, Toulouse.
- ◇ **February 2021 – October 2021:** INRIA Postdoctoral Fellow under the supervision of [Mario Sigalotti](#) and [Nastassia Pouradier Duteil](#), *Laboratoire Jacques-Louis Lions*, Paris.
- ◇ **November 2019 – February 2021:** CNRS Postdoctoral Fellow under the supervision of [Hélène Frankowska](#), *Institut de Mathématiques de Jussieu - Paris Rive Gauche*, Paris.
- ◇ **October 2016 – October 2019:** Ph.D. Student in Applied Mathematics under the supervision of [Francesco Rossi](#) and [Maxime Hauray](#), *Laboratoire d'Informatique et Systèmes*, Marseille & *Università degli Studi di Padova*, Padova.

Education

- ◇ **October 2016 – October 2019:** Ph.D. in Applied Mathematics. Specialisation in Control Theory.
 - **Title:** OPTIMAL CONTROL IN WASSERSTEIN SPACES
 - **Advisers:** [Francesco Rossi](#) (Director), *Università degli Studi di Padova*, Padova.
[Maxime Hauray](#) (Codirector), *Aix-Marseille Université*, Marseille.
 - **Jury:** [Filippo Santambrogio](#) (President), *Université Claude Bernard*, Lyon.
[Pierre Cardaliaguet](#) (Referee), *Université Paris-Dauphine*, Paris.
[Nicola Gigli](#) (Referee), *Scuola Internazionale Superiore Degli Studi Avanzati*, Trieste.
[José Antonio Carrillo](#) (Examinator), *Oxford University*, Oxford.
[Hélène Frankowska](#) (Examinator), *CNRS & Institut de Mathématiques de Jussieu*, Paris.
[Francesca Chittaro](#) (Examinator), *Laboratoire d'Informatique et Systèmes*, Toulon.

[Francesco Rossi](#) (Director), *Università degli Studi di Padova*, Padova.

[Maxime Hauray](#) (Codirector), *Aix-Marseille Université*, Marseille.

[Jean-Paul Gauthier](#) (Invited), *Laboratoire d'Informatique et Systèmes*, Toulon.

- ◇ **September 2015 – August 2016:** M.Sc. in Applied Mathematics. Specialisation in Optimisation, Calculus of Variations and Geometric Control. *Université Paris-Saclay*, Orsay.
- ◇ **September 2013 – September 2016:** Engineering curriculum in Applied Mathematics. Specialisation in Optimisation, Control Theory and Operational Research. *École Nationale Supérieure de Techniques Avancées (ENSTA Paris)*, Palaiseau.
- ◇ **September 2011 – September 2013:** French “Classes Préparatoires aux Grandes Écoles” with Mathematics and Physics majors (MPSI - MP*). *Lycée Blaise Pascal*, Orsay.
- ◇ **September 2008 – September 2011:** High School with Mathematics and Physics majors (1^{ère} - Terminale S). *Lycée Descartes*, Antony.

Community services

- ◇ **January 2024 – Now:** Coorganiser of the *Séminaire d'Automatique du Plateau de Saclay*.
- ◇ **July 2023 – Now:** Elected member of the *Conseil Scientifique d'Institut* (CSI) of the CNRS National Institute of Mathematics and its Interactions (INSMI) – Electoral College B1.

Grants

- ◇ **February 2024 – December 2024:** Functional support of 3000€ granted by the H-Code interdisciplinary object operating at the level of Université Paris-Saclay.
- ◇ **February 2021 – October 2021:** 15-month competitive Postdoctoral Fellowship from INRIA (interrupted to take my position at CNRS), *Université Pierre et Marie Curie*, Paris.
- ◇ **October 2016 – October 2019:** 3-year Ph.D. funding from the ARCHIMÈDE French Excellence Laboratory, *Laboratoire d'Informatique et Systèmes*, Marseille.

Conference organisations

- [O2] CIRM Workshop *Variational Analysis, Models and Methods in Measure Spaces – Control, Optimisation and Learning in Large-Scale Systems* (with [Massimo Fornasier](#), [Hélène Frankowska](#) and [Giuseppe Savaré](#)) – CENTRE INTERNATIONAL DE RENCONTRES MATHÉMATIQUES (CIRM), Marseille (April 29th - May 3rd 2024).
- [O1] Mini-course *Measure differential equations: modelling and numerical solutions* (with [Didier Henrion](#), [Swann Marx](#) and [Francesco Rossi](#)) – 22ND SYMPOSIUM ON MATHEMATICAL THEORY OF NETWORKS AND SYSTEMS (MTNS2022), Bayreuth (September 14th 2022).

Publications

The preprints of my articles can be found on my [Homepage](#) or via my [Google Scholar](#) account.

Submitted and under-revision

- [S2] B. Bonnet-Weill and M. Korda. [Set-Valued Koopman Theory for Control Systems](#). *Submitted*, 2024.

Published and accepted journal papers

- [J13] B. Bonnet-Weill and H. Frankowska. [Carathéodory Theory and A Priori Estimates for Continuity Inclusions in the Space of Probability Measures](#). *Nonlinear Analysis, Theory, Methods and Applications*, 247:113595, 2024.
- [J12] B. Bonnet-Weill and H. Frankowska. [On the Viability and Invariance of Proper Sets for Continuity Inclusions in Wasserstein Spaces](#). *SIAM Journal on Mathematical Analysis*, 56(3):2863-2914, 2024.
- [J11] R. Bonalli and B. Bonnet. [First-Order Pontryagin Optimality Conditions for Risk-Averse Stochastic Optimal Control Problems](#). *SIAM Journal on Control and Optimization*, 61(3):1881-1909, 2023.
- [J10] B. Bonnet, C. Cipriani, M. Fornasier and H. Huang. [A Measure Theoretical Approach to the Mean-Field Maximum Principle for Training NeurODEs](#). *Nonlinear Analysis*, 227:113161, 2023.
- [J9] B. Bonnet, N. Pouradier Duteil and M. Sigalotti. [Consensus Formation in First-Order Graphon Models with Time-Varying Topologies](#). *Mathematical Models and Methods in Applied Sciences*, 32(11):2121-2188, 2022.
- [J8] B. Bonnet and H. Frankowska. [Semiconcavity and Sensitivity Analysis in Mean-Field Optimal Control and Applications](#). *Journal de Mathématiques Pures et Appliquées*, 157:282-345, 2022.
- [J7] B. Bonnet and H. Frankowska. [Necessary Optimality Conditions for Optimal Control Problems in Wasserstein Spaces](#). *Applied Mathematics and Optimization*, 84:1281-1330, 2021.
- [J6] B. Bonnet and F. Rossi. [Intrinsic Lipschitz Regularity of Mean-Field Optimal Controls](#). *SIAM Journal on Control and Optimization*, 59(3):2011-2046, 2021.
- [J5] B. Bonnet and É. Flayac. [Consensus and Flocking Under Communication Failures for a Class of Cucker-Smale Systems](#). *Systems and Control Letters*, 152:104930, 2021.
- [J4] B. Bonnet and H. Frankowska. [Differential Inclusions in Wasserstein Spaces: The Cauchy-Lipschitz Framework](#). *Journal of Differential Equations*, 271:594-637, 2021.
- [J3] B. Bonnet. [A Pontryagin Maximum Principle in Wasserstein Spaces for Constrained Optimal Control Problems](#). *ESAIM COCV*, 25(52), 2019.
- [J2] B. Bonnet, J.P. Gauthier and F. Rossi. [Generic Singularities of the 3D-Contact Conjugate Locus](#). *Comptes Rendus Mathématiques*, 357(6):520-527, 2019.
- [J1] B. Bonnet and F. Rossi. [The Pontryagin Maximum Principle in the Wasserstein Space](#). *Calculus of Variations and Partial Differential Equations* 58:11, 2019.

Conference proceedings

- [C6] B. Bonnet-Weill and M. Sigalotti. [Exponential Consensus Formation in Time-Varying Multiagent Systems via Compactification Methods](#). *To appear in 2024 63rd IEEE Conference on Decision and Control (CDC)*, 2024.
- [C5] B. Bonnet and H. Frankowska. [Viability and Exponentially Stable Trajectories for Differential Inclusions in Wasserstein Spaces](#). *2022 61st IEEE Conference on Decision and Control (CDC)*, 5086-5091, 2022.
- [C4] B. Bonnet and H. Frankowska. [On the Properties of the Value Function Associated to a Mean-Field Optimal Control Problem of Bolza Type](#). *2021 IEEE Conference on Decision and Control (CDC)*, 4558-4563, 2021.
- [C3] B. Bonnet and F. Rossi. [Variance Optimization and Control Regularity for Mean-Field Dynamics](#). *IFAC-PapersOnLine*, 54 (19):13-18, 2021.

- [C2] B. Bonnet and H. Frankowska. [Mean-Field Optimal Control of Continuity Equations and Differential Inclusions](#). *2020 IEEE Conference on Decision and Control (CDC)*, 470-475, 2020.
- [C1] B. Bonnet and F. Rossi. [Sparse Control of Kinetic Cooperative Systems to Approximate Alignment](#). *Proceedings of the 20th IFAC World Congress*, 2017.

Presentations at conferences and seminars

Invited talks at conferences, seminars and workshops

- [I22] *A Set-Valued Approach to Koopman Operators for Control Systems* – DEPARTMENT OF MATHEMATICS AND CENTER FOR COMPLEX SYSTEMS, University of Namur (March 2024).
- [I21] *Measure Dynamics and Macroscopic Approximations of Multi-Agent Systems* – FÉDÉRATION DE MATHÉMATIQUES CENTRALESUPÉLEC, Plateau de Saclay (December 2023).
- [I20] *Set-Valued Dynamics with State-Constraints in Wasserstein Spaces* – WORKSHOP ON OPTIMAL TRANSPORT, MEAN-FIELD MODELS, AND MACHINE LEARNING, Technische Universität München (April 2023).
- [I19] *Set-Valued Generalisations of Koopman Operators* – WORKSHOP ON REAL ALGEBRAIC GEOMETRY WITH A VIEW TOWARDS KOOPMAN OPERATOR METHODS, Oberwolfach (March 2023).
- [I18] *New Results on Asymptotic Consensus Formation in Graphon Dynamics* – SÉMINAIRE D'AUTOMATIQUE DU PLATEAU DE SACLAY, Laboratoire des Signaux et Systèmes, Centrale Supélec, Gif-sur-Yvette (February 2023).
- [I17] *On the Lipschitz Regularity of Mean-Field Optimal Controls* – GROUPE DE TRAVAIL CONTRÔLE, Laboratoire Jacques-Louis Lions, Sorbonne-Université, Paris (January 2023).
- [I16] *Pontryagin Optimality Conditions in Wasserstein Spaces and their Application to the Training of NeurODEs* – SÉMINAIRE D'ANALYSE NON LINÉAIRE ET D'OPTIMISATION, Avignon Université, Avignon (October 2022).
- [I15] *Some Results Related to Consensus Formation in Graphon Dynamics* – CONFERENCE “ROUND MEAN-FIELD: CROWDS, OPINIONS, CELLS”, LYSM, Roma (September 2022).
- [I14] *When HJB Meets Pontryagin in Mean-Field Control* – INVITED SESSION “OPTIMAL CONTROL AND CALCULUS OF VARIATIONS ON METRIC SPACES”, 15th Viennese Conference on Optimal Control and Dynamics Games, Vienna (July 2022)
- [I13] *A Mean-Field Optimal Control Approach to Deep Learning* – INVITED SESSION “CONTRÔLE ET JEUX À CHAMP-MOYEN”, Journées SMAI MODE, Limoges (June 2022).
- [I12] *Consensus Formation, Macroscopic Approximations, and their Interactions in the context of Multi-Agent Dynamics* – DO SEMINAR, LAAS-CNRS, Toulouse (May 2022).
- [I11] *Set-Valued Dynamics in the Space of Probability Measures* – JOURNÉE RENCONTRE DE L'ÉQUIPE COMBINATOIRE ET OPTIMISATION, IMJ-PRG, Paris (April 2022).
- [I10] *A Mean-Field Optimal Control Approach to the Training of NeurODEs* – BRAINPOP SEMINAR, LAAS-CNRS, Toulouse (January 2022).
- [I9] *Fine Properties of the Value Function in Mean-Field Optimal Control* – INVITED SESSION “MEAN-FIELD GAMES AND APPLICATIONS”, PGMO Days, Palaiseau (December 2021).
- [I8] *Nonsmooth and Set-Valued Analysis in Wasserstein Spaces with Applications in Mean-Field Control* – SÉMINAIRE PARISIEN D'OPTIMISATION, IHP, Paris (November 2021).

- [17] *Sufficient Conditions for the Lipschitz Regularity of Mean-Field Optimal Controls* – GROUPE DE TRAVAIL DE CALCUL DES VARIATIONS, Remote talk (March 2021).
- [16] *Exponential Flocking under Communication Failures for some Cucker-Smale Models* – SEMINAR OF THE INRIA TEAM MAMBA, LJLL, Remote talk (March 2021).
- [15] *Continuity Inclusions and Applications in Mean-Field Optimal Control* – SEMINAR OF ANALYSIS AND APPLICATIONS, Université de Bretagne Occidentale, Remote talk (February 2021).
- [14] *Flocking for the Cucker-Smale Systems under Communication Failures* – SEMINAR OF INRIA TEAM CAGE, LJLL, Remote talk (May 2020).
- [13] *Intrinsic Lipschitz Regularity in Mean-Field Optimal Control Problems* – SEMINAR OF PROBABILITY, STATISTICS AND CONTROL THEORY, ENSTA Paris, Palaiseau (October 2019).
- [12] *Topics in Analysis and Optimal Control of Multi-Agent Systems* – SEMINARIO DI EQUAZIONI DIFFERENZIALE, Università degli Studi di Padova, Padova (March 2019).
- [11] *Optimal Control Problems in Wasserstein Spaces* – INVITED SESSION “VARIATIONAL ANALYSIS AND OPTIMAL CONTROL”, 14th Viennese Conference on Optimal Control and Dynamics Games, Vienna (August 2018).

Presentations at international conferences and research schools

- [P10] *Exponential Consensus Formation via Compactification Methods* – 63RD IEEE CONFERENCE ON DECISION AND CONTROL, Milano (December 2024).
- [P9] *A Meanfield Control Perspective on the Training of ResNets* – FRENCH-GERMAN-SPANISH CONFERENCE ON OPTIMIZATION, Gijon (June 2024).
- [P8] *Set-Valued Koopman Theory for Control Systems* – 2022 INTERNATIONAL SYMPOSIUM ON NONLINEAR THEORY AND ITS APPLICATIONS, Remote talk (December 2022).
- [P7] *Macroscopic Approximations of Multi-Agent Systems: An Introduction to Continuity Equations* – MINICOURSE ON MEASURE DIFFERENTIAL EQUATIONS, 25th International Symposium on Mathematical Theory of Networks and Systems, Bayreuth (September 2022).
- [P6] *Variance Optimization and Control Regularity in Mean-Fields Dynamics* – 7TH IFAC WORKSHOP ON LAGRANGIAN AND HAMILTONIAN METHODS FOR NONLINEAR CONTROL, Remote talk (October 2021).
- [P5] *Mean-Field Control and Continuity Inclusions* – 59TH IEEE CONFERENCE ON DECISION AND CONTROL, Remote talk (December 2020).
- [P4] *Some Problems in Modelling and Optimal Control of Multi-Agent Systems* – Poster session at the conference CROWDS: MODELS AND CONTROL, CIRM, Marseille (June 2019).
- [P3] *Optimal Control of Multi-Agent Systems: A Pontryagin Approach* – TOULOUSE WINTER SCHOOL IN CALCULUS OF VARIATIONS AND PROBABILITY THEORY, IMT, Toulouse (February 2019).
- [P2] *Optimal Control Problems in Wasserstein Spaces* – 12TH INTERNATIONAL YOUNG RESEARCHER WORKSHOP ON GEOMETRY, MECHANICS AND CONTROL, Università degli Studi di Padova, Padova (January 2018).
- [P1] *Sparse Alignment of Kinetic Cooperative Systems* – 20TH IFAC WORLD CONGRESS, Toulouse (July 2017).

Editorial activities

Reviewer for the journals *Probability Theory and Related Fields*, *SIAM Journal on Control and Optimization*, *SIAM Journal on Mathematics of Data Sciences*, *Journal of Differential Equations*, *Mathematics of Computations*, *Applied Mathematics and Optimization*, *Automatica*, *IEEE Transactions on Automatic Control*, *Journal of Mathematical Analysis and Applications*, *Journal of Dynamical and Control Systems*, as well as for the proceedings of the *IEEE Conference on Decision and Control*, *American Control Conference* and *IFAC World Congress*.

Teaching activities

- ◇ **2023 – Now:** Lectures for the course MEASURE THEORY AND LEBESGUE INTEGRATION.
Bachelor 3 level (specialisation), 27-hour teaching, *ENSTA Paris*, Palaiseau.
- ◇ **2019 – 2023:** Exercises sessions for the course DIFFERENTIABLE OPTIMISATION I.
Master 1 level, 15-hour teaching, *ENSTA Paris & UPSAY*, Palaiseau.
- ◇ **2020 – 2021:** Exercises sessions for the course DIFFERENTIABLE OPTIMISATION II.
Master 1 level, 15-hour teaching, *ENSTA Paris & UPSAY*, Palaiseau.
- ◇ **2020 – 2021:** Exercises sessions for the course OPTIMISATION.
Bachelor 3 level, 18-hour teaching, *UPP1*, Paris.
- ◇ **2019 – 2020:** Exercises sessions for the course QUADRATIC OPTIMISATION.
Bachelor 3 level, 15-hour teaching, *ENSTA Paris*, Palaiseau.
- ◇ **2017 – 2019:** Lectures for the course INTRODUCTION TO LEBESGUE INTEGRATION.
Bachelor 3 level, 4-hour teaching, *ECM*, Marseille.
- ◇ **2017 – 2019:** Lectures for the course INTRODUCTION TO OPTIMISATION THEORY.
Bachelor 3 level, 2-hour teaching, *ECM*, Marseille.
- ◇ **2017 – 2018:** Exercises sessions for the course PRELIMINARIES AND RECALLS IN OPTIMISATION.
Master 2 level, 14-hour teaching, *ECM & AMU*, Marseille.

Miscellaneous Skills, Hobbies and Interests

- ◇ **Languages:**
 - French (Mother tongue)
 - Italian (Basic, lived in Italy for a while)
 - English (Fluent, C2-level CEFR)
 - German & Chinese (Small remnants)
- ◇ **Hobbies:**
 - Drums (15-year regular practice)
 - Billiard (6-year regular practice)
 - Chess (2-year somewhat practice)
 - Bodhran (celtic traditional drums) & Guitar
 - Boulderling (indoor climbing) & bike travels
 - Video games, board games & card games
- ◇ **Interests:**
 - Epistemology of mathematics & physics
 - Sociology & political philosophy
 - Science fiction & fantasy novels
 - Climate sciences (MyCO2 ambassador)
 - Music in general
 - “Bandes dessinées” and mangas